

Duality

A linear map $V \rightarrow F$ turns each vector into a single number. The i -th coordinate in a basis does this, and so do evaluating a polynomial at a point, integrating a function, and taking the trace of a matrix. These **linear functionals** are the measurements we can make on a vector space, and together they form a vector space of their own, the dual space V^* .

The dual space mirrors V with every inclusion reversed. A subspace corresponds to the functionals that vanish on it. An injective map corresponds to a surjective one. The kernel of a map corresponds to the image of its dual map, whose matrix is the transpose. Passing to the dual turns a hard question into an easy one surprisingly often. It also explains facts we already know from a new angle: why a subspace of F^n is always the solution set of a homogeneous system, and why row rank equals column rank.

What you need. From Chapter 1, bases, dimension and complements (Definition 1.5.1, Definition 1.6.1, Theorem 1.7.6), and bases of infinite-dimensional spaces (Theorem 1.8.2) for the remarks on infinite dimension. From Chapter 2, the row space and null space of a matrix (Definition 2.5.2, Definition 2.5.3) and inverses by row reduction (Theorem 2.4.6). From Chapter 3, linear maps and the space $\mathcal{L}(V, W)$ (Definition 3.1.1, Theorem 3.4.1), the Rank–Nullity Theorem (Theorem 3.3.1), the matrix of a map and change of basis (Definition 3.6.1, Theorem 3.7.1), and quotient spaces with the First Isomorphism Theorem (Definition 3.9.3, Theorem 3.9.7).

4.-1.1 Roadmap

- **Linear functionals and the dual space.** Measurements as vectors; dual bases; separating vectors by functionals.
- **Annihilators.** The functionals that vanish on a subspace, and the formula $\dim U + \dim U^0 = \dim V$.
- **The double dual.** A natural isomorphism $V \rightarrow V^{**}$ that needs no basis, and what “natural” means.
- **The dual map and the transpose.** Pulling functionals back along a map; its matrix is the transpose; kernels and images trade places.
- **Duality for subspaces and quotients.** The dual of a subspace is a quotient of the dual, and the dual of a quotient is an annihilator; hyperplanes and codimension.
- **Hyperplanes and linear systems revisited.** Every subspace is cut out by equations; converting between parametric and implicit descriptions.

4.-1.2 Named moves

- **Test against functionals.** To show a vector is zero, show that every functional (or every vector of a dual basis) sends it to zero.
- **Pass to the dual and count.** Replace a subspace by its annihilator and use $\dim U + \dim U^0 = \dim V$.
- **Natural or chosen?** Ask whether an isomorphism needs a basis. $V \cong V^*$ does; $V \cong V^{**}$ does not.
- **Dualize a statement.** Injective becomes surjective, kernel becomes image, a matrix becomes its transpose, and inclusions reverse.